

PHYS505 Recitation — Feb 23, 2026

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Problem 4.1

Find at least 2 sets of non-0 moments for

- (a) $\pm q$ at $(x, y, z) = (\pm a, 0, 0)$ and $(0, \pm a, 0)$
- (b) $+q$ at $(x, y, z) = (0, 0, \pm a)$, $-2q$ at the origin
- (c) Using the lowest order term in the multipole expansion, plot the potential for part (b) in the x - y plane as a function of r for $r > a$
- (d) Compare (c) to the exact potential from Coulomb's law

Using the definition

$$q_{lm} = \int Y_{lm}^*(\theta', \phi') (r')^l \rho(\vec{x}') d^3x'$$

- (a) The charge density is made up of terms with δ -function factors only

$$\rho(x, y, z) = q \{ [\delta(x - a) - \delta(x + a)] \delta(y) + \delta(x) [\delta(y - a) - \delta(y + a)] \} \delta(z)$$

and evaluating the integral for the q_{lm} 's means simply substituting the appropriate coordinate values in the integrand. That is, for any function $f(x, y, z)$,

$$\int f(x, y, z) \rho(\vec{x}) d^3x = q \{ [f(a, 0, 0) - f(-a, 0, 0)] + [f(0, a, 0) - f(0, -a, 0)] \}$$

The monopole moment is 0

$$Y_{00}^* = \frac{1}{\sqrt{4\pi}} \qquad q_{00} = 0$$

There are a couple of finite dipole moments,

$$\begin{aligned} rY_{10}^* &= \sqrt{\frac{3}{4\pi}} z & q_{10} &= 0 \\ rY_{11}^* &= -\sqrt{\frac{3}{8\pi}} (x - iy) & \frac{q_{11}}{qa} &= -\sqrt{\frac{3}{2\pi}} (1 - i) \\ rY_{1,-1}^* &= \sqrt{\frac{3}{8\pi}} (x + iy) & \frac{q_{1,-1}}{qa} &= \sqrt{\frac{3}{2\pi}} (1 + i) \end{aligned}$$

Note that the usual definition of the dipole moment vector would be

$$\begin{aligned}
\vec{p} &= \int \vec{x} \rho(\vec{x}) d^3x = q \{ [a \hat{x} - (-a \hat{x})] + [a \hat{y} - (-a \hat{y})] \} \\
&= 2qa (\hat{x} + \hat{y}) \\
p_x &= -\sqrt{\frac{2\pi}{3}} (q_{11} - q_{1,-1}) = 2qa \\
p_y &= -i\sqrt{\frac{2\pi}{3}} (q_{11} + q_{1,-1}) = 2qa \\
p_z &= \sqrt{\frac{4\pi}{3}} q_{10} = 0
\end{aligned}$$

The quadrupole moments all vanish,

$$\begin{aligned}
r^2 Y_{20}^* &= \sqrt{\frac{5}{4\pi}} \frac{1}{2} (2z^2 - x^2 - y^2) & q_{20} &= 0 \\
r^2 Y_{21}^* &= -\sqrt{\frac{15}{8\pi}} z (x - iy) & q_{21} &= 0 = q_{2,-1} \\
r^2 Y_{22}^* &= \frac{1}{4} \sqrt{\frac{15}{2\pi}} (x^2 - y^2 - i2xy) & q_{22} &= 0 = q_{2,-2}
\end{aligned}$$

This is also different from the usual definition of the (traceless) quadrupole moment tensor but the components of that are also 0 since they are linear combinations of the q_{2m} 's.

$$\begin{aligned}
Q_{ij} &= \int (3x_i x_j - r^2 \delta_{ij}) \rho(x) d^3x \\
Q_{xx} &= \int (3x^2 - r^2) \rho(x) d^3x = \int (2x^2 - y^2 - z^2) \rho(x) d^3x = 0 \\
Q_{xy} &= \int 3xy \rho(x) d^3x = 0 \\
Q_{yy} &= \int (3y^2 - r^2) \rho(x) d^3x = \int (2y^2 - x^2 - z^2) \rho(x) d^3x = 0 \\
Q_{xz} &= \int 3xz \rho(x) d^3x = 0 \\
Q_{yz} &= \int 3xy \rho(x) d^3x = 0 \\
Q_{zz} &= \int (3z^2 - r^2) \rho(x) d^3x = \int (2z^2 - x^2 - y^2) \rho(x) d^3x = 0 = \sqrt{\frac{5}{4\pi}} \frac{1}{2} q_{30}
\end{aligned}$$

The sextupole moments are

$$\begin{aligned}
r^3 Y_{30}^* &= \sqrt{\frac{7}{4\pi}} \frac{z}{2} (2z^2 - 3x^2 - 3y^2) & q_{30} &= 0 \\
r^3 Y_{31}^* &= -\frac{1}{4} \sqrt{\frac{21}{4\pi}} (4z^2 - x^2 - y^2) (x - iy) & \frac{q_{31}}{qa^3} &= \frac{1}{2} \sqrt{\frac{21}{4\pi}} (1 - i) \\
r^3 Y_{3,-1}^* &= \frac{1}{4} \sqrt{\frac{21}{4\pi}} (4z^2 - x^2 - y^2) (x + iy) & \frac{q_{3,-1}}{qa^3} &= -\frac{1}{2} \sqrt{\frac{21}{4\pi}} (1 + i) \\
r^3 Y_{32}^* &= \frac{1}{4} \sqrt{\frac{105}{2\pi}} z (x^2 - y^2 - i2xy) & q_{32} &= 0 = q_{3,-2} \\
r^3 Y_{33}^* &= -\frac{1}{4} \sqrt{\frac{35}{4\pi}} (x^3 - i3x^2y - 3xy^2 + iy^3) & \frac{q_{33}}{qa^3} &= -\frac{1}{2} \sqrt{\frac{35}{4\pi}} (1 + i)
\end{aligned}$$

For higher order moments, note that all of these include integrals that are of the form

$$\int dx \int dy \int dz (x^j y^k z^n) \rho(x, y, z)$$

and with the given charge density, the only non-zero contributions come from

- no factors of z ($n = 0$)
- odd powers of x coupled with no factors of y or vice versa ($j = 0$ and odd k , or $k = 0$ and odd j)

and the value is proportional to a^{p+1} where p is the non-zero exponent. Also

- $r^l Y_{lm}$ is a homogeneous polynomial – the exponents in each term add up to l
- $r^l Y_{lm}$ is proportional to $(x \pm iy)^m$
- the remaining factor (polynomial of order $l - m$) includes powers of z and $x^2 + y^2$

This means

- all the $m = 0$ moments are 0 for $l > 0$ since there are no combinations with only odd powers of x , y and no z ;
- all the moments for even l are 0 for the same reason;
- for odd l , the $m = \pm l$ component is proportional to $(x \pm iy)^l$ and therefore has two non-zero contributions from the x^l and y^l terms and they don't cancel because one is real while the other is imaginary;
- similarly, other odd- m moments are also non-zero.

Having worked out the moments for $l = 0$ to 3 above, let's mention the next two orders

- all the octupole moments ($l = 4$) are 0
- the decapole moments ($l = 5$) with even m vanish, $q_{5,0} = q_{5,\pm 2} = q_{5,\pm 4} = 0$, while $q_{5,\pm 5}$, $q_{5,\pm 3}$, and $q_{5,\pm 1}$ are finite and proportional to qa^5

(b) The charge density in this case is

$$\rho(x, y, z) = q \delta(x) \delta(y) [\delta(z - a) + \delta(z + a) - 2\delta(z)]$$

The charge at the origin does not contribute to any of the moments except $l = 0$. Considerations similar to part (a) show that the only non-zero contributions come from combinations with no powers of x or y , that is, $m = 0$, and even powers of z , meaning even l . Note that for a point charge Q on the z -axis at $z = \zeta$,

$$\begin{aligned} \rho_Q(\vec{x}) &= Q \delta(x) \delta(y) \delta(z - \zeta) \\ Y_{l0} &= \sqrt{\frac{2l+1}{4\pi}} P_l\left(\frac{z}{r}\right) \\ q_{l0} &= \sqrt{\frac{2l+1}{4\pi}} P_l(1) \int z^l \rho_Q(\vec{x}) d^3x = \sqrt{\frac{2l+1}{4\pi}} Q \zeta^l \end{aligned}$$

Working out the lowest orders

$$\begin{aligned} q_{00} &= 0 & q_{20} &= \sqrt{\frac{5}{\pi}} qa^2 \\ q_{40} &= \frac{3}{\sqrt{\pi}} qa^4 & q_{60} &= \sqrt{\frac{13}{\pi}} qa^6 \end{aligned}$$

(c) The multipole expansion for the potential is

$$\begin{aligned} \Phi_b &= \frac{1}{\epsilon_0} \sum_{l,m} \frac{q_{lm}}{2l+1} \frac{Y_{lm}}{r^{l+1}} \\ \epsilon_0 \Phi_b &= \frac{q_{20}}{5} \sqrt{\frac{5}{4\pi}} \frac{1}{2} \left[3 \left(\frac{z}{r}\right)^2 - 1 \right] \frac{1}{r^3} + \frac{q_{40}}{9} \sqrt{\frac{9}{4\pi}} \frac{1}{8} \left[35 \left(\frac{z}{r}\right)^4 - 30 \left(\frac{z}{r}\right)^2 + 3 \right] \frac{1}{r^5} \\ &\quad + \frac{q_{60}}{13} \sqrt{\frac{13}{4\pi}} \frac{1}{16} \left[231 \left(\frac{z}{r}\right)^6 - 315 \left(\frac{z}{r}\right)^4 + 105 \left(\frac{z}{r}\right)^2 - 5 \right] \frac{1}{r^7} + \dots \\ &= \frac{q}{4\pi a} \left(\frac{a}{r}\right)^3 \left\{ \left[3 \left(\frac{z}{r}\right)^2 - 1 \right] + \left(\frac{a}{r}\right)^2 \left[35 \left(\frac{z}{r}\right)^4 - 30 \left(\frac{z}{r}\right)^2 + 3 \right] \right. \\ &\quad \left. + \left(\frac{a}{r}\right)^4 \left[231 \left(\frac{z}{r}\right)^6 - 315 \left(\frac{z}{r}\right)^4 + 105 \left(\frac{z}{r}\right)^2 - 5 \right] + \dots \right\} \end{aligned}$$

showing the three lowest order non-zero terms. On the midplane ($z = 0$, $r = s$), this reduces to

$$\Phi_b(z = 0, r) = -\frac{q}{4\pi\epsilon_0 a} \left(\frac{a}{r}\right)^3 \left[1 - \frac{3}{4} \left(\frac{a}{r}\right)^2 + \frac{5}{8} \left(\frac{a}{r}\right)^4 \right] + \mathcal{O} \left[\left(\frac{a}{r}\right)^9 \right]$$

(d) The exact potential can be expressed in cylindrical coordinates (with $s^2 = x^2 + y^2$) as

$$\begin{aligned}\Phi_b(s, z) &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{s^2 + (z-a)^2}} + \frac{1}{\sqrt{s^2 + (z+a)^2}} - \frac{2}{\sqrt{s^2 + z^2}} \right) \\ &= \frac{q}{4\pi\epsilon_0 a} \frac{a}{s} \left[\frac{1}{\sqrt{1 + \left(\frac{z-a}{s}\right)^2}} + \frac{1}{\sqrt{1 + \left(\frac{z+a}{s}\right)^2}} - \frac{2}{\sqrt{1 + \left(\frac{z}{s}\right)^2}} \right]\end{aligned}$$

On the midplane,

$$\begin{aligned}\Phi_b(r, z=0) &= \frac{q}{2\pi\epsilon_0 a} \frac{a}{r} \left[\frac{1}{\sqrt{1 + \left(\frac{a}{r}\right)^2}} - 1 \right] \\ &= \frac{q}{2\pi\epsilon_0 r} \left[-\frac{1}{2} \left(\frac{a}{r}\right)^2 + \frac{3}{8} \left(\frac{a}{r}\right)^4 - \frac{5}{16} \left(\frac{a}{r}\right)^6 + \dots \right] \\ &= -\frac{q}{4\pi\epsilon_0 a} \left(\frac{a}{r}\right)^3 \left[1 - \frac{3}{4} \left(\frac{a}{r}\right)^2 + \frac{5}{8} \left(\frac{a}{r}\right)^4 + \dots \right]\end{aligned}$$

which shows that if we only take the lowest order (quadrupole) term in the multipole expansion, the relative error is of order $(a/r)^2$ and therefore becomes negligible beyond a few a of distance from the origin. The plot below shows the multipole approximations (taking up to 3 terms) together with the exact solution.

